

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 01

DATE : 04-05-2019

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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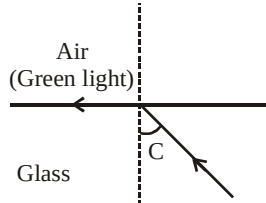
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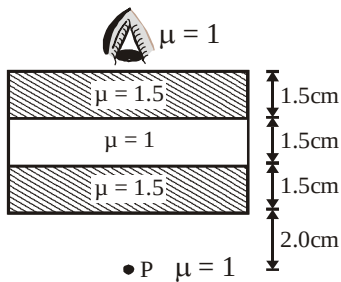
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BEWARE OF NEGATIVE MARKING

- A person can not see the objects beyond 50 cm. The power of a lens to correct this vision will be-
(1) +2D (2) -2D
(3) +5D (4) 0.5 D
- An achromatic convergent doublet of two lenses in contact has a power of + 2D. The convex lens has a power + 5D. What is the ratio of the dispersive powers of the convergent and divergent lenses
(1) 2 : 5 (2) 3 : 5 (3) 5 : 2 (4) 5 : 3
- A plano convex lens ($\mu = 1.5$) has radius of curvature 10 cm. It is silvered on its plane surface. Find focal length after silvering:-
(1) 10 cm (2) 20 cm
(3) 15 cm (4) 25 cm
- An equiconvex lens has a power of 5 dioptre. If it is made of glass of refractive index 1.5. then radius of curvature of its each surface will be ?
(1) 20 cm (2) 10 cm
(3) 40 cm (4) ∞
- Two lenses of power +2.50 D and -3.75 D are combined to form a compound lens. Its focal length in cm will be -
(1) 40 (2) -40
(3) -80 (4) 160
- Calculate the dispersive power for crown glass when $\mu_v = 1.60$ & $\mu_r = 1.50$:-
(1) 0.3 (2) 0.18
(3) 0.4 (4) None
- A point object is placed at the center of a glass sphere of radius 6 cm and refractive index 1.5. The distance of virtual image from the surface is :-
(1) 6 cm (2) 4 cm
(3) 12 cm (4) 9 cm
- A beam of white light is incident on glass-air interface from glass to air such that green light just suffers total internal reflection. The colors of the light which will come out to air are :-

(1) Violet, Indigo, Blue
(2) All colors except green
(3) Yellow, Orange, Red
(4) White light
- A diverging lens with unequal radii of curvature made of glass ($\mu_g = 1.5$) has a focal length of 40 cm. If it is immersed in a liquid of refractive index $\mu_l = 2$, then :-
(1) it behaves like a converging lens of 80 cm focal length
(2) it behaves like a converging lens of 20 cm focal length
(3) its focal length becomes 60 cm
(4) nothing can be said

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10. The image of point P when viewed from top of the slabs will be :-



- (1) 2.0 cm above P
(2) 1.5 cm above P
(3) 2.0 cm below P
(4) 1 cm above P
11. The velocity of light in a medium is half its velocity in air. If a ray of light emerges from such a medium into air, then find critical angle of incidence :-
(1) 15° (2) 30°
(3) 45° (4) 60°
12. In a thin prism of glass (refractive index 1.5), which of the following relations between the angle of minimum deviations δ_m and angle of refraction r of prism will be correct ?
(1) $\delta_m = r$ (2) $\delta_m = 1.5 r$
(3) $\delta_m = 2r$ (4) $\delta_m = \frac{r}{2}$
13. A real object is placed in front of a concave lens of focal length f at a distance equal to f from the lens. Then, the image is formed at :-
(1) infinity
(2) a distance $3f$
(3) a distance $f/2$
(4) a distance $2f$

14. A converging lens has a focal length of 1 m. The minimum distance between a real object and its real image formed by this lens is :-

(1) 0.5 m (2) 1 m
(3) 2 m (4) 4 m

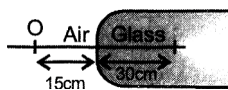
15. How will the image formed by a convex lens be affected, if the central portion of the lens is wrapped in black paper, as shown in Fig. ?



- (1) No image will be formed
(2) Full image will be formed but it is less bright
(3) Full image will be formed but without the central portion
(4) Two images will be formed, one due to each exposed half
16. A light ray strikes a glass surface from air at an angle of incidence 60° . The reflected ray and refracted ray are mutually perpendicular. Find the refractive index of glass :-
(1) 1.3 (2) 1.53 (3) 1.73 (4) 1.4
17. The colour are characterised by which of following character of light-
(1) Frequency
(2) Amplitude
(3) Wavelength
(4) Velocity

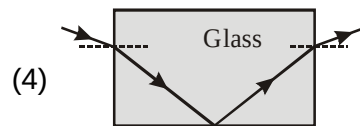
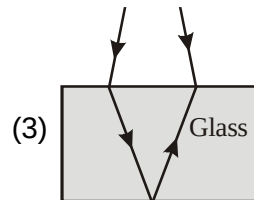
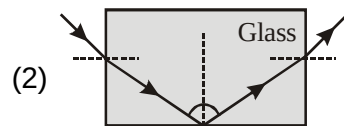
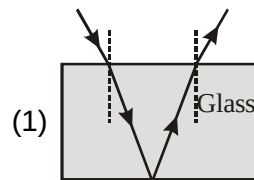
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18. A tall man of height 6 feet, want to see his full image. Then required minimum length of the mirror will be :-
 (1) 12 feet (2) 3 feet
 (3) 6 feet (4) Any length
19. The focal length of a spherical mirror is :-
 (1) Maximum for red light
 (2) Maximum for blue light
 (3) Maximum for white light
 (4) Same for all lights
20. Velocity of light in glass, whose refractive index w.r.t. air is 1.5, is 2×10^8 m/Sec. In a certain liquid, the velocity of light is found to be 2.5×10^8 m/sec. The refractive index of liquid w.r.t. air is
 (1) 0.64 (2) 0.80
 (3) 1.20 (4) 1.44
21. A thin Prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin Prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of Prism P_2 is
 (1) 5.33° (2) 4°
 (3) 3° (4) 2.6°
22. A point object O is placed in front of a glass rod having spherical end of radius of curvature 30 cm. The image that would formed be



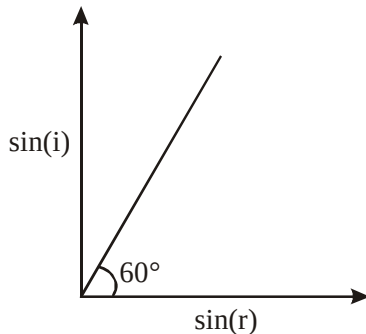
- (1) 30 cm left
 (2) infinity
 (3) 1 cm to the right
 (4) 18 cm to the left

23. An astronomical telescope has a magnifying power 10. The focal length of eyepiece is 20 cm. The focal length of objective is
 (1) 2 cm (2) 200cm
 (3) $\frac{1}{2}$ cm (4) $\frac{1}{200}$ cm
24. A ray of light passes through a rectangular glass block placed in a homogeneous medium. It is refracted and totally reflected. Which diagram shows a possible path of this ray ?

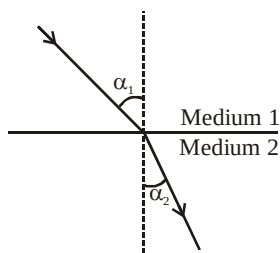


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25. A ray of light is incident on a medium with angle of incidence i and refracted into a second medium with angle of refraction r . The graph of $\sin(i)$ vs $\sin(r)$ is as shown in Fig. Then, the velocity of light in the first medium is n times the velocity of light in the second medium. What should be the value of n ?

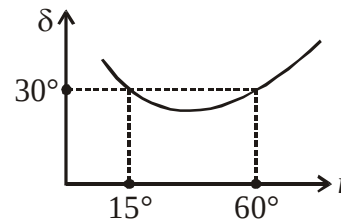


- (1) $\sqrt{3}$ (2) $1/\sqrt{3}$
(3) $\sqrt{3}/2$ (4) $2/\sqrt{3}$
26. A beam of light propagates through a medium 1 and falls onto another medium 2, at an angle α_1 as shown in Fig. After that, it propagates in medium 2 at an angle α_2 as shown. The light's wavelength in medium 1 is λ_1 . What is the wavelength of light in medium 2 ?



- (1) $\frac{\sin \alpha_1}{\sin \alpha_2}$ (2) $\frac{\sin \alpha_2}{\sin \alpha_1} \lambda_1$
(3) $\frac{\cos \alpha_1}{\cos \alpha_2} \lambda_1$ (4) $\frac{\cos \alpha_2}{\cos \alpha_1} \lambda_1$

27. Figure shows the graph of angle of deviation δ versus angle of incidence i for a light ray striking a prism. The prism angle is :-



- (1) 30° (2) 45°
(3) 60° (4) 75°
28. A ray of light passes from glass, having a refractive index of 1.6, to air. The angle of incidence for which the angle of refraction is twice the angle of incidence is :-

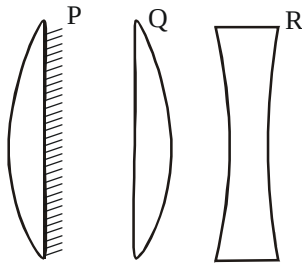
- (1) $\sin^{-1}\left(\frac{4}{5}\right)$
(2) $\sin^{-1}\left(\frac{3}{5}\right)$
(3) $\sin^{-1}\left(\frac{5}{8}\right)$
(4) $\sin^{-1}\left(\frac{2}{5}\right)$

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29. Two convex lenses of powers $4D$ and $6D$ are separated by a distance of $1/6$ m. The power of the optical system so formed is :-

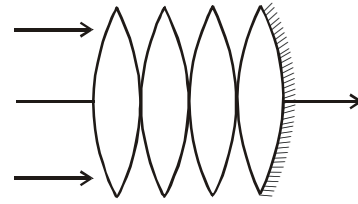
(1) $-6D$
(2) $+6D$
(3) $10D$
(4) $2D$

30. Figure shows three lenses of equal radii of curvature of the curved surfaces and equal refractive index. The ratio of focal lengths of P, Q and R is :-



(1) $1 : 1 : 1$
(2) $1 : 1 : -1$
(3) $-1 : 2 : 1$
(4) $-1 : 2 : -1$

31. Four identical lenses are kept one beside the other on the same optical axis as shown in the figure. The right surface of rightmost lens is silvered. Consider each surface of curvature 20 cm. The focal length of the combined system is $[\mu_{\text{lens}} = 1.5]$:-



(1) 2cm (2) -2cm
(3) 5cm (4) -5cm

32. The dispersive power of crown and flint glasses are 0.02 and 0.04 , respectively. An achromatic converging lens of focal length 40 cm is made by keeping two lenses, one of crown glass and the other of flint glass, in contact with each other. The focal lengths of the two lenses are:-

(1) 20 cm and 40 cm
(2) 20 cm and -40 cm
(3) -20 cm and 40 cm
(4) 10 cm and -20 cm

33. If an equilateral prism minimum deviation is 30° , what will be the angle of incidence ?

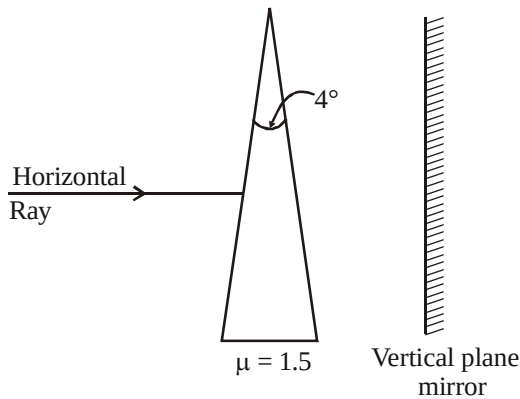
(1) 30° (2) 60°
(3) 45° (4) 90°

34. A spherical mirror forms an image of magnification 3 . The object distance, if focal length of mirror is 24 cm, may be :-

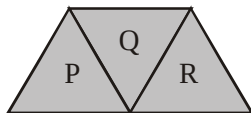
(1) 32 cm, 24 cm
(2) 32 cm, 16 cm
(3) 32 cm only
(4) 16 cm only

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35. For the situations shown in the figure, determine the angle by which the mirror should be rotated, so that the light ray will retrace its path after refraction through the prism and reflection from the mirror ?

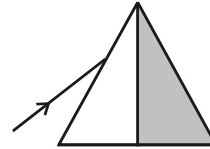


- (1) 1° ACW
(2) 1° CW
(3) 2° ACW
(4) 2° CW
36. A given ray of light suffers minimum deviation in an equilateral prism P. Additional prisms Q and R of identical shape and material are now added to P as shown in the figure. The ray will suffer :-

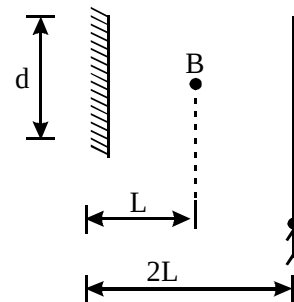


- (1) Greater deviation
(2) Same deviation
(3) No deviation
(4) Total internal reflection

37. A light ray is incident upon a prism in minimum deviation position and suffers a deviation of 34° . If the shaded half of the prism is knocked off, the ray will :-



- (1) Suffer a deviation of 34°
(2) Suffer a deviation of 68°
(3) Suffer a deviation of 17°
(4) Not come out of the prism
38. Far points of myopic eye is 250 cm, then the focal length of the lens to be used will be :-
- (1) -250 cm (2) $-250/9$ cm
(3) + 250 cm (4) $+ 250/9$ cm
39. A point source of light B is placed at a distance L in front of the center of a mirror of width 'd' hung vertically on a wall. A man walks in front of the mirror along a line parallel to the mirror at a distance 2L from it as shown in figure. The greatest distance over which he can see the image of the light source in the mirror is :-



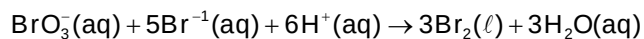
- (1) $d/2$ (2) d (3) 2d (4) 3d

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| <p>40. A spherical surface of radius of curvature R separates air (refractive index 1.0) from glass (refractive index 1.5). The centre of curvature is in the glass. A point object P placed in air is found to have a real image Q in the glass. The line PQ cuts the surface at a point O, and $PO = OQ$. The distance PO is equal to :-</p> <p>(1) $5R$ (2) $3R$
 (3) $2R$ (4) $1.5 R$</p> <p>41. Two beams of red and violet colors are made to pass separately through a prism (angle of the prism is 60°). In the position of minimum deviation, the angle of refraction will be :-</p> <p>(1) 30° for both the colors
 (2) greater for the violet color
 (3) greater for the red color
 (4) equal but not 30° for both the colors</p> <p>42. A ray of light travelling in water is incident on its surface open to air. The angle of incidence is θ, which is less than the critical angle. Then, there will be :-</p> <p>(1) only a reflected ray and no refracted ray
 (2) only a refracted ray and no reflected ray
 (3) a reflected ray and a refracted ray and the angle between them would be less than $180^\circ - 2\theta$
 (4) a reflected ray and a refracted ray and the angle between them would be greater than $180^\circ - 2\theta$</p> | <p>43. The magnification produced by an astronomical telescope for normal adjustment is 10 and the length of telescope is 1.1m. The magnification when image is formed at least distance of distinct vision is :-</p> <p>(1) 14
 (2) 6
 (3) 16
 (4) 18</p> <p>44. A simple telescope, consisting of an objective of focal length 60 cm and a single eye lens of focal length 5 cm is focussed on a distant object in such a way that parallel rays come out from the eye lens. If the object subtends an angle 2° at the objective, the angular width of the image :-</p> <p>(1) 10°
 (2) 24°
 (3) 50°
 (4) $1/6^\circ$</p> <p>45. An eye specialist prescribes spectacles having combination of convex lens of focal length 40 cm in contact with a concave lens of focal length 25 cm. The power of this lens combination in diopters is :-</p> <p>(1) +1.5
 (2) -1.5
 (3) +6.67
 (4) -6.67</p> |
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46. In the following reaction, how is the rate of appearance of the underlined product related to the rate of disappearance of the underlined reactant



(1) $(-)\frac{\Delta[\text{BrO}_3^-]}{\Delta t} = \frac{\Delta[\text{Br}_2]}{\Delta t}$

(2) $(-)\frac{1}{3}\frac{\Delta[\text{BrO}_3^-]}{\Delta t} = \frac{\Delta[\text{Br}_2]}{\Delta t}$

(3) $(-)\frac{\Delta[\text{BrO}_3^-]}{\Delta t} = \frac{1}{3}\frac{\Delta[\text{Br}_2]}{\Delta t}$

(4) None of these

47. For $\text{A}_{(\text{g})} + \text{B}_{(\text{g})} \rightarrow \text{C}_{(\text{g})}$; rate = $k[\text{A}]^{1/2}[\text{B}]^2$, if initial concentration of A and B are increased by factor of 4 and 2 respectively, then the initial rate is changed by the factor

- (1) 4 (2) 6
(3) 8 (4) None of these

48. A first order reaction is 75% completed in 100 minutes. How long will it take for it's 87.5% completion?

- (1) 125 min (2) 150 min
(3) 175 min (4) 200 min

49. The rate constant for a first order reaction which has half life 480 sec

- (1) $1.44 \times 10^{-3} \text{ sec}^{-1}$
(2) $1.44 \times 10^{-2} \text{ sec}^{-1}$
(3) $0.72 \times 10^{-3} \text{ sec}^{-3}$
(4) $2.88 \times 10^{-3} \text{ sec}^{-3}$

50. The activation energy of the reaction, $\text{A} + \text{B} \rightarrow \text{C} + \text{D} + 38 \text{ kcal}$ is 20 kcal, what would be the activation energy of the reaction, $\text{C} + \text{D} \rightarrow \text{A} + \text{B}$

- (1) 20 kcal (2) - 20 kcal
(3) 18 kcal (4) 58 kcal

51. In a catalytic reaction involving the formation of ammonia by Haber's process $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ the rate of appearance of NH_3 was measured as $2.5 \times 10^{-4} \text{ mole L}^{-1} \text{ s}^{-1}$ The rate of disappearance of H_2 will be -

- (1) $2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
(2) $1.25 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
(3) $3.75 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$
(4) $5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$

52. Rate of formation of SO_3 according to the reaction $2\text{SO}_2 + \text{O}_2 \rightarrow 2\text{SO}_3$ is $1.6 \times 10^{-3} \text{ kg min}^{-1}$ Hence rate at which SO_2 reacts is :-

- (1) $1.6 \times 10^{-3} \text{ kg min}^{-1}$
(2) $8.0 \times 10^{-4} \text{ kg min}^{-1}$
(3) $3.2 \times 10^{-3} \text{ kg min}^{-1}$
(4) $1.28 \times 10^{-3} \text{ kg min}^{-1}$

53. Which of the following statement is not correct for the reaction: $4\text{A} + \text{B} \rightarrow 2\text{C} + 2\text{D}$

- (1) The rate of disappearance of B is twice the rate of appearance of C
(2) The rate of disappearance of B is one fourth the rate of disappearance of A
(3) The rate of formation of D is one half the rate of consumption of A
(4) The rate of formation of C and D are equal

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54. For gaseous reaction, rate = $k [A] [B]$. If

volume of container is reduced to $\frac{1}{4}$ of initial, then the rate of the reaction will be times of initial :-

- (1) $\frac{1}{8}$ (2) 8 (3) $\frac{1}{16}$ (4) 16

55. The rate of certain hypothetical reaction $A + B + C \rightarrow$ products is given

by $r = \frac{-d[A]}{dt} = K[A]^{\frac{1}{2}}[B]^{\frac{1}{3}}[C]^{\frac{1}{4}}$. The order of the reaction -

- (1) 1 (2) $\frac{1}{2}$ (3) 2 (4) $\frac{13}{12}$

56. For the reaction $H_2(g) + Br_2(g) \rightarrow 2HBr(g)$, the experimental data suggests, Rate = $K [H_2] [Br_2]^{\frac{1}{2}}$. The order for this reaction is

- (1) 2 (2) $1\frac{1}{2}$ (3) 1 (4) $2\frac{1}{2}$

57. In a certain gaseous reaction between X and Y, $X + 3Y \rightarrow XY_3$. The initial rates are reported as follows -

[X]	[Y]	Rate
0.1 M	0.1 M	0.002 Ms^{-1}
0.2 M	0.1 M	0.002 Ms^{-1}
0.3 M	0.2 M	0.008 Ms^{-1}
0.4 M	0.3 M	0.018 Ms^{-1}

The rate law is -

- (1) $r = K[X][Y]^3$ (2) $r = K[X]^0[Y]^2$
(3) $r = K[X][Y]$ (4) $r = [X]^0[Y]^3$

58. Calculate the order of the reaction in A and B :

[A] (mol L ⁻¹)	[B] (mol L ⁻¹)	Rate of reaction (mol L ⁻¹ min ⁻¹)
0.01	0.01	1.0×10^{-4}
0.01	0.03	9.0×10^{-4}
0.03	0.03	2.70×10^{-3}

- (1) 1 and 0 (2) 1 and 2
(3) 0 and 1 (4) None

59. Time required to complete a half fraction of a reaction varies inversely to the concentration of reactant then the order of reaction is -

- (1) Zero (2) 1
(3) 2 (4) 3

60. The rate law for the single step reaction $2A + B \rightarrow 2C$, is given by -

- (1) Rate = $K[A][B]$
(2) Rate = $K[A]^2[B]$
(3) Rate = $K[2A][B]$
(4) Rate = $K[A]^2[B]^0$

61. For an elementary process $2X + Y \rightarrow Z + W$, the molecularity is -

- (1) 2 (2) 1
(3) 3 (4) Unpredictable

62. For the reaction $A + B \rightarrow$ products, it is found that the order of A is 1 and the order

of B is $\frac{1}{2}$. When the conc., of both A and B

are increased four times, the rate will increase by a factor of :-

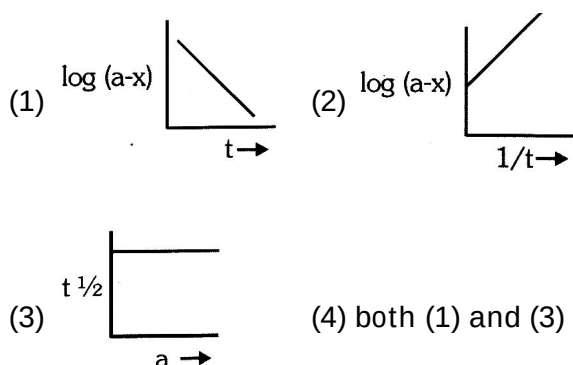
- (1) 16 (2) 8
(3) 6 (4) 4

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63. Following mechanism has been proposed for a reaction $2A + B \rightarrow D + E$
 $A + B \rightarrow C + D$ (slow)
 $A + C \rightarrow E$ (fast)
 The rate law expression for the reaction is -
 (1) $r = K[A]^2 [B]$
 (2) $r = K[A] [B]$
 (3) $r = K[A]^2$
 (4) $r = K[A][C]$
64. For reaction $NO_2 + CO \rightarrow CO_2 + NO$, the rate expression is, $\text{Rate} = K [NO_2]^2$ The number of molecules of CO involved in the slowest step will be -
 (1) 0 (2) 1 (3) 2 (4) 3
65. The acid hydrolysis of ester is -
 (1) 1 order reaction
 (2) Bimolecular reaction
 (3) Pseudo unimolecular reaction
 (4) All
66. The reaction $2A + B \rightarrow \text{products}$, follows the mechanism
 $2A \rightleftharpoons A_2$ (fast)
 $A_2 + B \rightarrow P$ (slow) The order of the reaction is -
 (1) 1.5 (2) 3 (3) 1 (4) 2
67. K for a zero order reaction is $2 \times 10^{-2} \text{ mol L}^{-1} \text{ sec}^{-1}$. If the concentration of the reactant after 25 sec is 0.5 M, the initial concentration must have been.
 (1) 0.5 M (2) 1.25 M
 (3) 12.5 M (4) 1.0 M
68. The decomposition of N_2O_5 occurs as, $2N_2O_5 \rightarrow 4NO_2 + O_2$, and follows first order kinetics; hence
 (1) The reaction is bimolecular
 (2) The reaction is unimolecular
 (3) $t_{1/2} \propto a^\circ$
 (4) None
69. 75 % of a first order reaction was found to complete in 32 min. When will 50 % of the same reaction complete -
 (1) 24 min (2) 16 min
 (3) 8 min (4) 4 min
70. A first order reaction has a half life period of 69.3 sec. At 0.10 mol L^{-1} reactant concentration, the rate will be -
 (1) $10^{-4} \text{ M sec}^{-1}$
 (2) $10^{-3} \text{ M sec}^{-1}$
 (3) $10^{-1} \text{ M sec}^{-1}$
 (4) $6.93 \times 10^{-1} \text{ M sec}^{-1}$
71. 99 % of a first order reaction was completed in 32 min. when will 99.9 % of the reaction complete ?
 (1) 50 min (2) 46 min
 (3) 49 min (4) 48 min
72. What is the half life of a radioactive substance if 87.5% of any given amount of the substance disintegrate in 40 minutes ?
 (1) 160 min (2) 10 min
 (3) 20 min (4) 13 min 20 sec.

SPACE FOR ROUGH WORK

73. The reaction $L \rightarrow M$ is started with 10 g/L. After 30 and 90 minute, 5 g and 1.25 g/L are left respectively. The order of reaction is
(1) 0 (2) 2 (3) 1 (4) 3
74. If doubling the initial concentration of a reactant doubles $t_{1/2}$ of the reaction, the order of the reaction is-
(1) 3 (2) 2 (3) 1 (4) 0
75. The rate constant for a reaction is $10.8 \times 10^{-5} \text{ mol L}^{-1} \text{ s}^{-1}$. The reaction obeys
(1) First order
(2) Zero order
(3) Second order
(4) All are wrong
76. The rate constant for a second order reaction is $8 \times 10^{-5} \text{ M}^{-1} \text{ min}^{-1}$. How long will it take a 1M solution to be reduced to 0.5 M.
(1) $8.665 \times 10^3 \text{ min}$ (2) $8 \times 10^{-3} \text{ min}$
(3) $1.25 \times 10^4 \text{ min}$ (4) $4 \times 10^5 \text{ min}$
77. In the case of first order reaction, the ratio of time required for 99.9 % completion to 50 % completion is :-
(1) 2 (2) 5 (3) 10 (4) None
78. Which of the following curves represents a 1st order reaction :-



79. The following data were obtained at a certain temperature for the decomposition of ammonia
- | p (mm) | 50 | 100 | 200 |
|-----------|------|------|------|
| $t_{1/2}$ | 3.64 | 1.82 | 0.91 |
- The order of the reaction is :-
(1) 0 (2) 1 (3) 2 (4) 3
80. The rate constants k_1 and k_2 for two different reactions are $10^{16} e^{-2000/T}$ and $10^{15} e^{-1000/T}$, respectively. The temperature at which $k_1 = k_2$ is
(1) $\frac{2000}{2.303} \text{ K}$ (2) 2000 K
(3) $\frac{1000}{2.303} \text{ K}$ (4) 1000K
81. For the reaction $A + B \rightarrow \text{products}$, it is observed that :-
(a) on doubling the initial concentration of A only, the rate of reaction is also doubled and
(b) on doubling the initial concentrations of both A and B, there is a change by a factor of 8 in the rate of the reaction.
The rate of this reaction is given by :-
(1) $\text{rate} = k[A][B]$
(2) $\text{rate} = k[A]^2[B]$
(3) $\text{rate} = k[A][B]^2$
(4) $\text{rate} = k[A]^2[B]^2$
82. Half life period of a first-order reaction is 1386 seconds. The specific rate constant of the reaction is :-
(1) $5.0 \times 10^{-2} \text{ s}^{-1}$ (2) $5.0 \times 10^{-3} \text{ s}^{-1}$
(3) $0.5 \times 10^{-5} \text{ s}^{-1}$ (4) $0.5 \times 10^{-3} \text{ s}^{-1}$

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83. In a zero-order reaction for every 10° rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C , the rate of the reaction will become:

- (1) 64 times (2) 128 times
(3) 256 times (4) 512 times

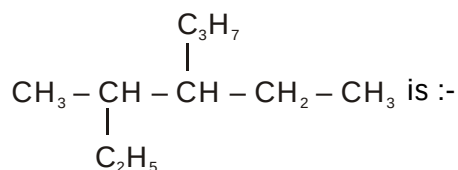
84. A reaction having equal energies of activation for forward and reverse reactions has :-

- (1) $\Delta H = \Delta G = \Delta S = 0$ (2) $\Delta S = 0$
(3) $\Delta G = 0$ (4) $\Delta H = 0$

85. On increasing temperature from 200K to 220K rate of reaction A increases by 3 times and rate of reaction B increases by 9 times then correct relationship between activation energy of A and B is :-

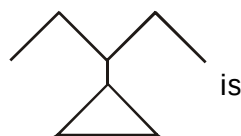
- (1) $E_A = 3E_B$ (2) $3E_A = E_B$
(3) $E_B = 2E_A$ (4) $E_A = 2E_B$

86. IUPAC name of the compound



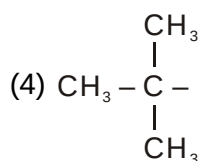
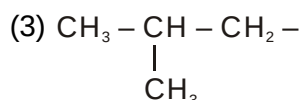
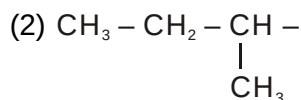
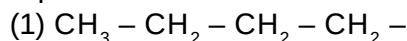
- (1) 2-Ethyl-3-propylpentane
(2) 4-Ethyl-3-methylheptane
(3) 4-Ethyl-3-propylpentane
(4) 3-methyl-4-ethylheptane

87. IUPAC name of the compound

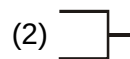
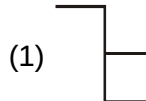


- (1) pentylcyclopropane
(2) 3-pentyl cyclopropane
(3) 3-cyclopropylpentane
(4) 1-Ethyl-1-cyclopropyl propane

88. Secondary butyl substituent is correctly represented as :-

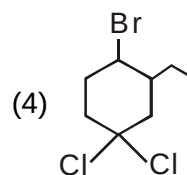
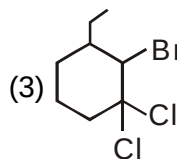
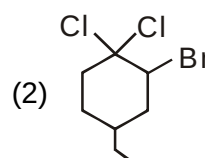
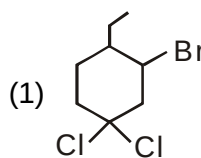


89. 3-methylpentane is IUPAC name of :-



(4) All of these

90. 3-Bromo-1,1-dichloro-4-ethylcyclohexane is correctly represented as :-



SPACE FOR ROUGH WORK

91. One of the following is almost not found in living organism?
 (1) Si (2) Mg
 (3) Ca (4) S
92. Lipid comes under acid insoluble fraction during analysis of the chemical composition of tissues. Why?
 (1) It has low molecular weight
 (2) It has high molecular weight
 (3) It is polymer
 (4) On grinding, the biomembranes are broken into pieces and form insoluble vesicles
93. The four elements called "Big-four" which make up 95% of all elements found in a living system are
 (1) C, H, O, N (2) C, H, O, P
 (3) C, H, O, S (4) C, N, O, P
94. Biomolecules are
 (1) inorganic materials
 (2) organic materials
 (3) all carbon compound obtained from living tissues
 (4) only DNA, RNA
95. Amino acid has a
 (1) substituted methane
 (2) substituted ethane
 (3) any acid having amino group
 (4) derivative of indolacetic acid
96. Protein amino acids are called
 (1) α -amino acids
 (2) β -amino acids
 (3) γ -amino acids
 (4) σ -amino acids
97. Which of the group in amino acids makes them unique?
 (1) R group
 (2) NH_2 group
 (3) COOH group
 (4) None of these

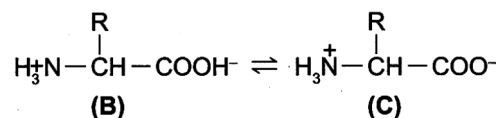
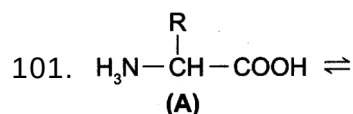
98. Based up on the number of NH_2 and COOH amino acids are

- (1) 3 types (2) 2 types
 (3) 4 types (4) 20 types

99. Which one is correct?

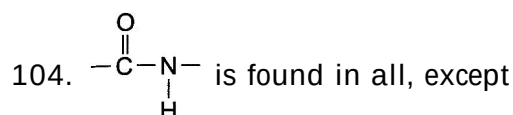
Acidic Amino Acids		Basic Amino Acids	Neutral Amino Acid
(1)	Glutamic acid	Lysine	Valine
(2)	Lysine	Valine	Glutamic acid
(3)	Glutamic acid	Valine	Lysine
(4)	Lysine	Glutamic acid	Valine

100. The aromatic amino acid (s) is (are)
 (1) Tyrosine (2) Phenylalanine
 (3) tryptophan (4) All of these



Which of the above is Zwitter ionic form?

- (1) A (2) C
 (3) B (4) All of these
102. Types of amino acids universally used in protein synthesis are
 (1) 20 (2) 80 (3) 64 (4) 61
103. Which one is the most abundant organic biomolecule in a cell?
 (1) Water (2) Protein
 (3) DNA (4) mRNA



- (1) polypeptides
 (2) enzymes
 (3) amino acids
 (4) proteins

105. Primary structure of protein is due to

- (1) Peptide bonds
- (2) Hydrogen bonds
- (3) -S-S bond
- (4) Ionic bonds

106. A β -pleated sheet organization in a polypeptide chain is an example of

- (1) 1° structure
- (2) 2° structure
- (3) 3° structure
- (4) 4° structure

107. Polysaccharides, polypeptides, and polynucleotides have in common that they all

- (1) contain amino acids
- (2) are formed in condensation reactions
- (3) contain nitrogen
- (4) come out in acid soluble pool

108. Which of the following statements concerning polymers is not true?

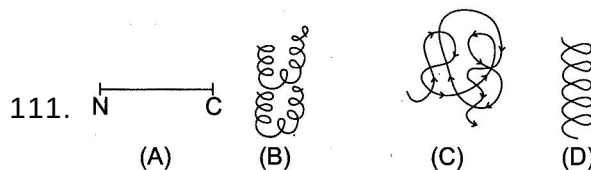
- (1) Polymers are synthesized from monomers during condensation
- (2) Polymers are synthesized from monomers by addition of water
- (3) Polymers consist of at least two types of monomers
- (4) Both (2) and (3)

109. Adult human haemoglobin consists of

- (1) 2 subunits (α , α)
- (2) 2 subunits (β , β)
- (3) 4 subunits (2α , 2β)
- (4) 3 subunits (2α , 1β)

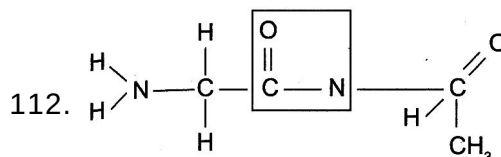
110. Which is the highest structural organization found in all enzymes?

- (1) 2°
- (2) 3°
- (3) 1°
- (4) 4°



What kinds of the structures of proteins are shown in the above figure?

- (1) A = 1° structure, B = 2° structure, C = 3° structure, D = 4° structure
- (2) A = 4° structure, B = 2° structure, C = 3° structure, D = 1° structure
- (3) A = 1° structure, B = 4° structure, C = 3° structure, D = 2° structure
- (4) A = 4° structure, B = 3° structure, C = 2° structure, D = 1° structure



What is the structure enclosed by a box?

- (1) Amino acid
- (2) Peptide bond
- (3) Glycosidic bond
- (4) Zwitter ion

113. Lipids are

- (1) insoluble in water
- (2) readily soluble in organic solvent
- (3) important constituent of biological membrane
- (4) all of these

114. Arachidic acid and palmitic acids have how many carbons in each of them?

- (1) 16, 16
- (2) 16, 20
- (3) 20, 16
- (4) 18, 18

115. Glycerol is

- (1) Tetrahydroxy propane
- (2) Trihydroxy propane
- (3) Trihydroxy butane
- (4) Tetrahydroxy butane

116. $\text{CH}_3 - (\text{CH}_2)_{14} - \text{COOH}$

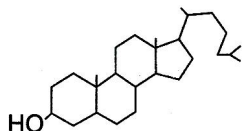
This is the formula of

- (1) Phospholipid
- (2) Palmitic acid
- (3) Triglyceride
- (4) Arachidonic acid

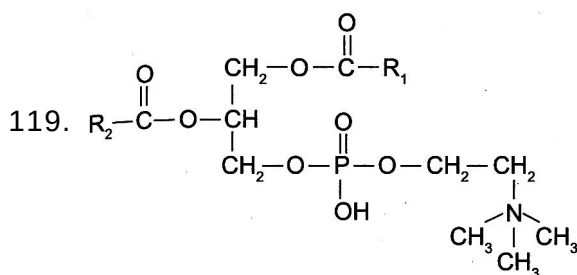
117. Lecithin is

- (1) Phospholipid
- (2) Carbohydrate
- (3) Protein
- (4) Amino acid

118. This molecule is related to



- (1) Phospholipid
- (2) Lecithin
- (3) Cholesterol
- (4) Oleic acid



The above diagrammatic representation is the formula of

- (1) Lecithin
- (2) Cholesterol
- (3) Uridylic acid
- (4) Phosphatic acid

120. Which one is secondary metabolite?

- (1) Amino acid
- (2) Lipid
- (3) Alkaloid
- (4) Protein

121. Which of the following secondary metabolites are used as drugs?

- (1) Abrin and Ricin
- (2) Vinblastin and Curcumin
- (3) Anthocyanine
- (4) Amino acid

122. Three important polysaccharides made up of glucose monomers are

- (1) Sucrose, lactose, and maltose
- (2) Chitin, glycogen, and starch
- (3) Starch, glycogen, and cellulose
- (4) RNA, DNA, and starch

123. In animals, glucose is stored as _____ while in plants it is stored as _____.

- (1) cellulose, starch
- (2) starch, glycogen
- (3) cellulose, glycogen
- (4) glycogen, Starch

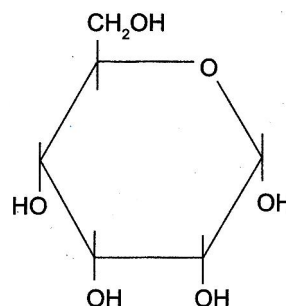
124. Cellulose is found in

- (1) cell wall plant
- (2) paper
- (3) cotton
- (4) All of these

125. Inulin consists of

- (1) only glucose
- (2) only fructose
- (4) only maltose
- (4) protein

126. The structural formula belongs to



- (1) Glucose
- (2) Ribose
- (3) Sucrose
- (4) Deoxyribose

127. (I) Right end of polysaccharide is called reducing end while left end is called non reducing end

(II) Starch can hold I_2 molecules in its helical secondary structure but cellulose being non-helical, cannot hold I_2

(III) Starch and glycogen are branched molecules

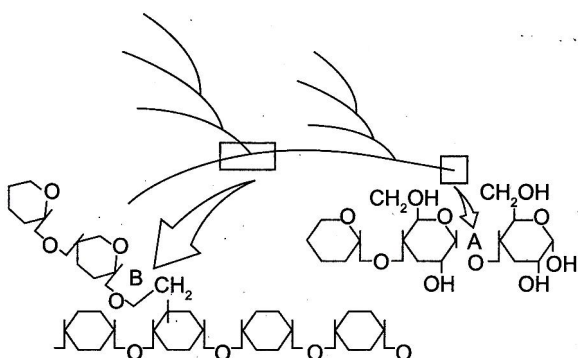
(IV) Starch in plant and glycogen in animals are store houses of energy

- (1) I and IV are correct
- (2) II and III are correct
- (3) Only IV is correct
- (4) All are correct

128. In polysaccharide, the individual monosaccharides are linked by a bond. This bond is formed by dehydration. The bond is

- (1) Ester bond
- (2) Glycosidic bond
- (3) H-bond
- (4) Ionic bond

129. Identify A and B bonds in the following diagrammatic representation of a portion of glycogen.



- (1) A = 1-4 α -glycosidic bonds, B = 1-6 α -glycosidic bonds
- (2) A = 1-6 α -glycosidic bonds, B = 1-4 α -glycosidic bonds
- (3) A = 1-1 α -glycosidic bonds, B = 1-1 α -glycosidic bonds
- (4) A = 1-4 α -glycosidic bonds, B = 1-4 α -glycosidic bonds

130. Which of the following fatty acids is liquid at room temperature?

- (1) Palmitic acid
- (2) Stearic acid
- (3) Oleic acid
- (4) Arachidic acid

131. Which one is the most abundant protein in the animal world?

- (1) Trypsin
- (2) Haemoglobin
- (3) Collagen
- (4) Insulin

132. Which one out of A-D given below correctly represents the structural formula of the basic amino acid?

A	B	C	D
$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{C}=\text{O} \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{OH} \end{array}$	$\begin{array}{c} \text{CH}_2\text{OH} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH}_2 \end{array}$	$\begin{array}{c} \text{NH}_2 \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{CH}_2 \\ \\ \text{NH}_2 \end{array}$

- (1) A (2) B (3) C (4) D

133. Which one of the following is a non-reducing carbohydrate?

- (1) Maltose
- (2) Sucrose
- (3) Lactose
- (4) Ribose 5-phosphate

134. Which one of the following statements is wrong?

- (1) Sucrose is a disaccharide
- (2) Cellulose is a polysaccharide
- (3) Uracil is a pyrimidine
- (4) Glycine is a sulphur containing amino acid

135. A typical fat molecule is made up of

- (1) three glycerol molecules and one fatty acid molecule
- (2) one glycerol and three fatty acid molecules
- (3) one glycerol and one fatty acid molecule
- (4) three glycerol and three fatty acid molecules

136. Each and every organism can live only for a certain period of time. The period from birth to the natural death of an organism represents its
- (1) Asexual reproduction
 - (2) Sexual reproduction
 - (3) Development
 - (4) Life span
137. Whatever be the life span, death of every individual organisms is a certainty, i.e., no individual is immortal, except
- (1) Human beings
 - (2) *Amoeba* and *Paramoecium*
 - (3) Single-celled organisms
 - (4) Both (2) and (3)
138. A biological process in which an organism gives rise to young ones (offspring) similar to itself is called
- (1) Reproduction
 - (2) Fertilisation
 - (3) Parthenogenesis
 - (4) Gametogenesis
139. In the life span of any organism there is a cycle of
- (1) Birth, growth and death
 - (2) Birth, fertilisation and death
 - (3) Juvenile, reproduction and senescence
 - (4) Pre-fertilisation, fertilisation and post-fertilisation
140. Which one enables the continuity of the species generation after generation?
- (1) Reproduction
 - (2) Fertilisation
 - (3) Life-cycle
 - (4) Life-span
141. When offspring is produced by a single parent with or without the involvement of gamete formation, the reproduction is called
- (1) Asexual
 - (2) Sexual
 - (3) Parthenogenetic
 - (4) Either (1) or (2)
142. In plants, the vegetative propagation such as runner, rhizome, sucker, tuber, offset, bulb are all capable of giving rise to new offspring. These structures are called -
- (1) Clones
 - (2) Grafts
 - (3) Vegetative propagules
 - (4) Adventitious buds
143. In potato, sugarcane, banana, ginger the new plantlets invariably arises from
- (1) The nodes present in the modified stems
 - (2) The nodes present in the modified roots
 - (3) The internodes present in the modified stems
 - (4) The margin of the leaves
144. Roots are used in vegetative propagation of
- (1) Ginger
 - (2) Sugar cane
 - (3) Sweet Potato
 - (4) Potato
145. Vegetative propagation in *Pistia* occurs by
- (1) Sucker
 - (2) Runner
 - (3) Offset
 - (4) Stolon
146. Which is wrongly matched?
- (1) *Agave*—bulbils
 - (2) *Penicillium*—conidia
 - (3) Water Hyacinth—runner
 - (4) *Bryophyllum*—leaf buds
147. Human gametes differ from all other body cells in being
- (1) Haploid
 - (2) Diploid
 - (3) Immotile
 - (4) Without cell wall
148. Process of sexual reproduction which involves meiosis and syngamy is
- (1) Apomixis
 - (2) Amphimixis
 - (3) Vegetative propagation
 - (4) Parthenogenesis

149. Gametogenesis refers to the process of
 (1) Fusion of two gametes
 (2) Fusion of two gametangia
 (3) Formation of two types of gametes
 (4) Formation of male gametes only

150. The motile reproductive structures of algae and fungi, which directly give rise to new individuals are called
 (1) Zygosporos (2) Zoospores
 (3) Cysts (4) Conidia

151. Which of the following pairs is not correctly matched?

Mode of reproduction	Example
(1) Rhizome	Banana
(2) Binary fission	<i>Sargassum</i>
(3) Conidia	<i>Penicillium</i>
(4) Offset	Water hyacinth

152. Which type of fertilization is found in human being?
 (1) External
 (2) Internal
 (3) Both (1) and (2)
 (4) None of the above

153. During favourable condition the encysted amoeba divides by multiple fission and produces pseudopodiospores. This phenomenon is known as
 (1) Budding
 (2) Sporulation
 (3) Fragmentation
 (4) Regeneration

154. The internal buds of fresh water sponges are called
 (1) Choanocyte (2) Gemmule
 (3) Osculum (4) Blastula

155. Binary fission is a type of
 (1) Vegetation propagation
 (2) Asexual reproduction
 (3) Sexual reproduction
 (4) Nuclear fragmentation

156. Potatoes are cultivated by

- (1) Seeds (2) Tuber
 (3) Buds on tubers (4) Bud

157. Clone is a group of individuals got through
 (1) Sexual reproduction
 (2) Cross pollination
 (3) Vegetative propagation
 (4) Hybridisation

158. A piece of Potato tuber will form a new plant if it has
 (1) Branches (2) Stored food
 (3) Roots (4) Scales/eyes

159. Match the columns I and II, and choose the correct combination from the options given.

- | Column I | Column II |
|--------------|---------------|
| 1. Coconut | a. Monoecious |
| 2. Papaya | b. Dioecious |
| 3. Tapeworm | |
| 4. Earthworm | |

- (1) a—1, b—2, a—3, b—4
 (2) b—1, a—2, b—3, a—4
 (3) a—1, b—2, a—3, a—4
 (4) a—1, b—2, b—3, b—4

160. Which one is found only in aquatic plant?
 (1) Runner (2) Stolon
 (3) Tuber (4) Offset

161. Eye of Potato is
 (1) Apical bud
 (2) Axillary bud
 (3) Accessory bud
 (4) Adventitious bud

162. Individuals of a clone have
 (1) Same age
 (2) Same height
 (3) Same genome
 (4) Same number of leaves

163. Asexually produced organism inheriting all the characters of the parent is
 (1) Offspring (2) Clone
 (3) Variety (4) Hybrid

- | | |
|--|--|
| <p>164. Type of asexual reproduction found in <i>Hydra</i> is</p> <ol style="list-style-type: none"> (1) Gemmule formation (2) Sporulation (3) Binary fission (4) Budding <p>165. A population of genetically identical individuals obtained from asexual reproduction is</p> <ol style="list-style-type: none"> (1) Callus (2) Clone (3) Deme (4) Aggregate <p>166. Estrous cycle is a characteristic of</p> <ol style="list-style-type: none"> (1) Human females (2) Mammalian females (3) Mammalian females other than primates (4) Mammals <p>167. Syngamy means</p> <ol style="list-style-type: none"> (1) Fusion of gametes (2) Fusion of cytoplasms (3) Fusion of two similar spores (4) Fusion of two dissimilar spores <p>168. Which one of the following characteristics is not shared by birds ?</p> <ol style="list-style-type: none"> (1) Viviparity (2) Warm blooded nature (3) Ossified endoskeleton (4) Breathing using lungs <p>169. In bryophytes and pteridophytes, transport of male gametes requires</p> <ol style="list-style-type: none"> (1) Birds (2) Water (3) Wind (4) Insects <p>170. In organism with haplontic life cycle, zygote divides by</p> <ol style="list-style-type: none"> (1) Mitosis to form haploid spores (2) Meiosis to form gametes (3) Mitosis to form gametes (4) Meiosis to form haploid spores <p>171. Every sexually reproducing organism, including human beings begin life as a single cell called</p> <ol style="list-style-type: none"> (1) Gamete (2) Spore (3) Embryo (4) Zygote | <p>172. The process of the development of embryo from the zygote is called</p> <ol style="list-style-type: none"> (1) Gametogenesis (2) Sporogenesis (3) Embryogenesis (4) Oogenesis <p>173. During embryogenesis, zygote undergoes</p> <ol style="list-style-type: none"> (1) Cell division (mitosis) (2) Cell differentiation (3) Meiosis/reduction division (4) Both (1) and (2) <p>174. Animals in which development of zygote takes place outside the body of female parent and they lay fertilised/ unfertilised egg are called</p> <ol style="list-style-type: none"> (1) Oviparous (2) Viviparous (3) Ovoviviparous (4) Marsupials <p>175. Animals in which development of zygote takes place inside the body of female parent, i.e., they give birth to young ones are called</p> <ol style="list-style-type: none"> (1) Oviparous (2) Viviparous (3) Ovoviviparous (4) Marsupials <p>176. Fertilised eggs covered by calcareous shells are found in</p> <ol style="list-style-type: none"> (1) Fishes and amphibians (2) Reptiles, birds and mammals (3) Amphibians, reptiles and birds (4) Reptiles and birds <p>177. Asexual reproduction does not involve the</p> <ol style="list-style-type: none"> (1) Syngamy (2) Fusion of gametes (3) Both (1) and (2) (4) None of the above |
|--|--|

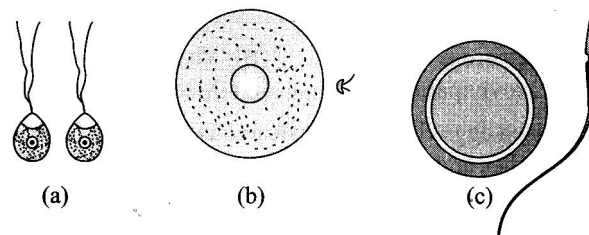
178. Which is incorrect about flowering plant?

- (1) After fertilisation the ovary develops into fruit and ovules develop into seed
- (2) The ovary wall after syngamy converted into pericarp which is protective in function
- (3) The zygote is formed inside the ovule
- (4) None of the above

179. External water is not required for fertilization of

- (1) Pteridophytes
- (2) Bryophytes
- (3) Thallophytes
- (4) Spermatophytes.

180. Recognise the figure and find out the correct matching.



- (1) a—heterogametes of *Cladophora*,
b—heterogametes of *Homo sapiens*,
c—isogametes of *Fucus*
- (2) a—isogametes of *Fucus*,
b—heterogametes of *Cladophora*,
c—heterogametes of humans
- (3) a—isogametes of *Cladophora*,
b—heterogametes of *Homo sapiens*,
c—heterogametes of *Fucus*
- (4) a—isogametes of *Cladophora*,
b—heterogametes of *Fucus*,
c—heterogametes of human beings

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